



Online reduced kernel principal component analysis for process monitoring



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ABSTRACT

Kernel principal component analysis (KPCA), which is a nonlinear extension of principal component analysis (PCA), has gained significant attention as a monitoring method for nonlinear processes. However, KPCA cannot perform well for dynamic systems and when the training data set is large. Therefore, in this paper, an online reduced KPCA algorithm for process monitoring is proposed. The process monitoring performances are studied using two examples: a numerical example and Tennessee Eastman Process (TEP). The simulation results demonstrate the effectiveness of the proposed method when compared to the online KPCA method.

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1. Introduction

The last century has seen an increasingly emergence of manufacturing and process industries, with the rise of energy costs and environmental regulations. In these environments, processes are highly automated. Therefore, the monitoring algorithms are important to detect any fault that might occur. Multivariate statistical methods such as principal component analysis (PCA) [8,16,3,27], independent component analysis (ICA) [17,20,10] and partial least squares (PLS) [28,1] have been applied for process monitoring. PCA is the most used technique because of the simplicity of its concept and implementation. It transforms the original input variables into variables in a low dimensional subspace. Process monitoring is detecting faults in a system. In the PCA method, fault detection is performed using fault detection indices. The most used indices are the squared prediction error (SPE) that monitors the residual subspace and the Hotelling T^2 statistic for monitoring the principal component subspace. The PCA method assumes that the relationships among variables are linear and may ignore important information in the behaviors of nonlinear systems. However, the

majority of industrial processes have nonlinear characteristics. To address this issue, many nonlinear PCA methods have been developed [18,6,26]. The most popular one is kernel principal component analysis (KPCA) [26,23] because of its simplicity and elegance. It has been shown in [21], that KPCA method is used for monitoring continuous processes and it provided better performance when compared to linear PCA method. KPCA can efficiently compute principal components in high-dimensional feature space using integral operators and nonlinear kernel functions. It consists in mapping measurements from their original space to a higher dimensional feature space where PCA is performed. Fault detection using KPCA [22,4,24,9] can be performed similarly to linear PCA using SPE and T^2 statistics in the feature space. In the KPCA method, the size of the kernel matrix is given by the number of the training samples. Every time a new sample x is collected, N kernel functions have to be evaluated to form the kernel vector $k(x)$. Therefore, using KPCA model for process monitoring imposes a high computational cost when the training data set is large. K-means clustering [25] was developed in order to reduce the amount of training data. It requires the number of clusters in advance, therefore using it for process monitoring leads to numerous errors because it doesn't consider the variation of parameters according to the process operation changes.

In [15], the fault detection singular value decomposition reduced kernel principal component analysis (SVD-RKPCA) method

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is proposed for online monitoring of nonlinear processes. The SVD-RKPCA method is performed in two phases (offline and online). It consists in finding a reduced data set by selecting the variables that have the highest projection variance to build the initial KPCA model in the offline phase and then updating this model using the SVD-KPCA technique in the online phase. In the SVD-RKPCA method, the dictionary size is fixed during the training and testing phases. Thus, the large size of reduced data set can lead to a significant reduction in the computational efficiency. In addition, when the size of reduced data set is small, the updated information on the systems can be ignored and SVD-RKPCA cannot perform well for dynamic systems. To overcome the problems stated above, a new method referred as online reduced KPCA for process monitoring is proposed. The method aims to reduce the number of kernel functions. The reduced set of kernel functions is called dictionary. A new criterion based on squared prediction error evaluation determines the relevance of discarding or adding the kernel function to the dictionary $\mathcal{D}_r$. The resulting dictionary contains linearly independent kernel functions. The proposed method is performed in the online phase and takes into consideration the dynamic behaviors of the systems by changing the model structure. In the proposed online reduced KPCA method, the size of dictionary is variable and depends on the evaluated criterion. The proposed method forms the dictionary in an online way and then it updates the old KPCA model and uses it for process monitoring. The new method reduces significantly the computation time required to detect faults in nonlinear systems while conserving the structure of the data in the feature space. The performance of the proposed method is evaluated using a numerical example and Tennessee Eastman Process (TEP). The simulation results show that the proposed method provides better detection performances when compared to the online KPCA method.

This paper is organized as follows: an overview of the PCA and KPCA methods is given in Section 2. The fault detection index SPE is presented in Section 3. Section 4 presents the proposed online reduced KPCA method. Then, in Section 5, the fault detection performance is studied using a numerical example and TEP. At the end, the conclusions are presented in Section 6.

2. Preliminaries

2.1. Principal component analysis

The main idea of the principal component analysis (PCA) method is discarding the noise and collinearity between process variables, while preserving the most important information of the original data set. A PCA model is established based on the data collected under normal operating conditions for process monitoring.

Let $X \in \mathbf{R}^{N \times m}$ be the data matrix with N measurements and m variables. For PCA modeling, the matrix X is scaled to zero mean and unit variance. The covariance matrix Σ of the data matrix X and its eigenvalue decomposition are given by:

$$\begin{aligned} \Sigma &= \frac{1}{N} X^T X \\ &= [\hat{P} \quad \tilde{P}] \begin{bmatrix} \hat{\Lambda} & 0 \\ 0 & \tilde{\Lambda} \end{bmatrix} [\hat{P} \quad \tilde{P}]^T \end{aligned} \quad (1)$$

where $\hat{P} \in \mathbf{R}^{m \times \ell}$ and $\tilde{P} \in \mathbf{R}^{m \times (m-\ell)}$, representing the first ℓ and last $(m - \ell)$ eigenvectors of Σ , respectively. ℓ is the number of retained principal components (PCs). The diagonal matrices $\hat{\Lambda}$ and $\tilde{\Lambda}$ contain the ℓ highest and $(m - \ell)$ lowest eigenvalues of Σ , respectively. The data matrix X can be decomposed as follows:

$$X = \hat{X} + \tilde{X} \quad (2)$$

where $\hat{X}$ and $\tilde{X}$ are the projection to the principal subspace (PS), spanned by the columns of $\hat{P}$, and to the residual subspace (RS) spanned by the columns of $\tilde{P}$.

2.2. Kernel principal component analysis

Since the classical PCA method performs well only on linear processes, a nonlinear PCA technique called kernel PCA (KPCA), has been developed by Schölkopf et al. [26] and widely used to model various nonlinear processes. The key idea of KPCA is to map first the input space into a feature space $\mathcal{H}$ via nonlinear mapping ϕ and then perform a linear PCA in $\mathcal{H}$. Let the normalized training set be $x_1, x_2, \dots, x_N \in \mathbf{R}^m$. The measured inputs are projected into the feature space using the mapping function ϕ :

$$\phi : x_i \in \mathbf{R}^m \mapsto \phi(x_i) \in \mathbf{R}^h \quad (3)$$

where $h \gg N$ is the dimension of the feature space. An important property of the feature space is that the dot product of two vectors $\phi(x_i)$ and $\phi(x_j)$, $i, j = 1, \dots, N$, can be calculated as:

$$\phi(x_i)^T \phi(x_j) = \langle \phi(x_i), \phi(x_j) \rangle = k(x_i, x_j) \quad (4)$$

where k is called the kernel function. There are several types of kernel functions. One of the most used kernel functions is the radial basis function (RBF) which is defined as:

$$k(x_i, x_j) = \exp \left[\frac{-\|x_i - x_j\|^2}{\sigma^2} \right] \quad (5)$$

where σ is the width of a Gaussian function. Assuming that the vectors in the feature space are scaled to zero mean, the mapped data in $\mathcal{H}$ is arranged as

$$\mathcal{X} = [\phi(x_1) \quad \phi(x_2) \quad \dots \quad \phi(x_N)]^T.$$

The covariance matrix Q in the feature space $\mathcal{H}$ can be calculated as:

$$\begin{aligned} Q &= \frac{1}{N-1} \mathcal{X}^T \mathcal{X} \\ &= \frac{1}{N-1} \sum_{j=1}^N \phi(x_j) \phi(x_j)^T \end{aligned} \quad (6)$$

where it is assumed that $\sum_{j=1}^N \phi(x_j) = 0$. Similarly to the linear PCA, KPCA in the feature space is equivalent to solving the following eigenvalue problem:

$$\begin{aligned} \mu_k V_k &= Q V_k \\ &= \frac{1}{N-1} \sum_{j=1}^N \langle \phi(x_j), V_k \rangle \phi(x_j) \end{aligned} \quad (7)$$

where μ_k and V_k represent respectively the k th eigenvalue and eigenvector of Q , and $\langle \cdot, \cdot \rangle$ is the dot product. For $\mu_k \neq 0$, it is clear from Eq. (7) that every eigenvector V_k of Q can be considered as a linear combination of $\phi(x_1), \dots, \phi(x_N)$. Thus, there are coefficients $\alpha_{k,i}$ ($i = 1, \dots, N$) such that:

$$\begin{aligned} V_k &= \sum_{i=1}^N \alpha_{k,i} \phi(x_i) \\ &= \mathcal{X}^T \alpha_k \end{aligned} \quad (8)$$

where $\alpha_k = [\alpha_{k,1} \ \alpha_{k,2} \ \dots \ \alpha_{k,N}]^T$. Then, by multiplying $\phi(x_k)$ with both sides of Eq. (7) and inserting Eq. (8) in Eq. (7), we get:

$$\mu_k \sum_{i=1}^N \alpha_{k,i} \langle \phi(x_k), \phi(x_i) \rangle = \frac{1}{N-1} \sum_{i=1}^N \alpha_{k,i} \langle \phi(x_k), \sum_{j=1}^N \phi(x_j) \rangle \langle \phi(x_j), \phi(x_i) \rangle \quad (9)$$

Using the kernel function $K_{ij} = k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$, Eq. (9) can be written as:

$$\mu_k \sum_{i=1}^N \alpha_{k,i} K_{k,i} = \frac{1}{N-1} \sum_{i=1}^N \alpha_{k,i} \sum_{j=1}^N K_{kj} K_{ji} \quad (10)$$

Therefore, the eigenvalue problem of Eq. (7) can be expressed as:

$$\lambda_k \alpha_k = K \alpha_k \quad (11)$$

where $\lambda_k = (N-1)\mu_k$, $K \in \mathbf{R}^{N \times N}$ is the kernel matrix which is defined by K_{ij} and $\alpha_k = [\alpha_{k,1} \ \alpha_{k,2} \ \dots \ \alpha_{k,N}]^T$. In the feature space $\mathcal{H}$, the eigenvector V_k form a matrix V_f which is defined as $V_f = [V_1 \ V_2 \ \dots \ V_\ell \ V_{\ell+1} \ \dots \ V_N]$, where ℓ is the number of retained principal components (PCs). The first part $\hat{P}_f = [V_1 \ V_2 \ \dots \ V_\ell] \in \mathbf{R}^{N \times \ell}$ and the last part $\tilde{P}_f = [V_{\ell+1} \ \dots \ V_N] \in \mathbf{R}^{N \times (N-\ell)}$ represent the eigenvectors defining respectively the principal subspace and the residual subspace. In order to guarantee the normality constraint of eigenvectors in $\mathcal{H}$, $\langle V_k, V_k \rangle = 1$, Eqs. (8) and (11) are used to derive:

$$\left\{ \begin{aligned} \langle V_k, V_k \rangle &= \left\langle \sum_{i=1}^N \alpha_{k,i} \phi(x_i), \sum_{j=1}^N \alpha_{k,j} \phi(x_j) \right\rangle \\ &= \sum_{i=1}^N \sum_{j=1}^N \alpha_{k,i} \alpha_{k,j} \langle \phi(x_i), \phi(x_j) \rangle \\ &= \sum_{i=1}^N \sum_{j=1}^N \alpha_{k,i} \alpha_{k,j} K_{ij} \\ &= \langle \alpha_k, K \alpha_k \rangle \\ &= \lambda_k \langle \alpha_k, \alpha_k \rangle \\ &= 1 \end{aligned} \right. \quad (12)$$

Therefore, α_k needs to have a norm of $1/\sqrt{\lambda_k}$. Let $\bar{\alpha}_k$ be the unit norm eigenvector corresponding to λ_k , such that:

$$\alpha_k = \frac{1}{\sqrt{\lambda_k}} \bar{\alpha}_k \quad k = 1, \dots, N \quad (13)$$

Using Eqs. (8) and (13), the matrix $\hat{P}_f$ can be written as:

$$\begin{aligned} \hat{P}_f &= [\lambda^T \bar{\alpha}_1 \lambda_1^{-(1/2)} \ \dots \ \lambda^T \bar{\alpha}_\ell \lambda_\ell^{-(1/2)}] \\ &= \lambda^T P \Lambda^{-(1/2)} \end{aligned} \quad (14)$$

where $P = [\bar{\alpha}_1 \ \bar{\alpha}_2 \ \dots \ \bar{\alpha}_\ell]$ and $\Lambda = \text{diag}(\lambda_1 \ \dots \ \lambda_\ell)$. The principal and residual components, respectively, $t \in \mathbf{R}^\ell$ and $\tilde{t} \in \mathbf{R}^{N-\ell}$ of a test vector x , are then extracted by projecting $\phi(x)$ into the principal and residual spaces, as follows:

$$\left\{ \begin{aligned} t &= \hat{P}_f^T \phi(x) \\ &= \Lambda^{-(1/2)} P^T k(x) \\ \tilde{t} &= \tilde{P}_f^T \phi(x) \end{aligned} \right. \quad (15)$$

where $k(x) = [k(x_1, x) \ k(x_2, x) \ \dots \ k(x_N, x)]^T$. In the feature space $\mathcal{H}$, the vector $\phi(x)$ has zero mean. If this is not the case, the

vectors in $\mathcal{H}$ have to be scaled to zero mean using the mean of the training data. The scaled vector $\bar{\phi}(x)$ is given by:

$$\begin{aligned} \bar{\phi}(x) &= \phi(x) - \frac{1}{N} \sum_{i=1}^N \phi(x_i) \\ &= \phi(x) - [\phi(x_1) \ \phi(x_2) \ \dots \ \phi(x_N)] I_N \end{aligned} \quad (16)$$

where $I_N = \frac{1}{N} [1 \ \dots \ 1]^T \in \mathbf{R}^N$. According to Eq. (16), the kernel function of two scaled vectors $\bar{\phi}(x_i)$ and $\bar{\phi}(x_j)$ is defined as:

$$\begin{aligned} \bar{k}(x_i, x_j) &= \bar{\phi}(x_i)^T \bar{\phi}(x_j) \\ &= k(x_i, x_j) - k^T(x_i) I_N - k^T(x_j) I_N + I_N^T K I_N \end{aligned} \quad (17)$$

Similarly, the scaled kernel vector $\bar{k}(x)$ is calculated as:

$$\left\{ \begin{aligned} \bar{k}(x) &= [\bar{\phi}(x_1) \ \dots \ \bar{\phi}(x_N)]^T \bar{\phi}(x) \\ &= F(k(x) - K I_N) \\ F &= I - E \end{aligned} \right. \quad (18)$$

where I is the identity matrix and $E \in \mathbf{R}^{N \times N}$ is a matrix with elements $1/N$. Finally, the centered kernel matrix $\bar{K}$ is calculated as:

$$\begin{aligned} \bar{K} &= [\bar{\phi}(x_1) \ \dots \ \bar{\phi}(x_N)]^T [\bar{\phi}(x_1) \ \dots \ \bar{\phi}(x_N)] \\ &= F K F \end{aligned} \quad (19)$$

Therefore, it is preferred to have centered vectors in the feature space. For the rest of the paper, we assume that we have centered vectors in the feature space.

3. Fault detection

In the feature space $\mathcal{H}$, the data are distributed linearly. Thus, the two statistics SPE and T^2 can be used for fault detection in $\mathcal{H}$ [22,4,24,9,25,14]. The squared prediction error SPE index is defined as the norm of the residual vector in the feature space $\mathcal{H}$ and it is expressed as:

$$\left\{ \begin{aligned} SPE &= \tilde{t}^T \tilde{t} \\ &= \phi(x)^T \tilde{P}_f \tilde{P}_f^T \phi(x) \\ &= \phi(x)^T \tilde{C}_f \phi(x) \\ &= \phi(x)^T (I - \hat{P}_f \hat{P}_f^T) \phi(x) \\ &= \phi(x)^T \phi(x) - \phi(x)^T \hat{P}_f \hat{P}_f^T \phi(x) \\ &= k(x, x) - k^T(x) C k(x) \end{aligned} \right. \quad (20)$$

where I is the identity matrix, $\tilde{C}_f = \tilde{P}_f \tilde{P}_f^T = I - \hat{P}_f \hat{P}_f^T$ is the projection matrix to the residual space and $C = P \Lambda^{-1} P^T$. The control limit of the SPE index is calculated from the χ^2 -distribution and it is defined as [22]:

$$SPE_{\alpha} = g \chi_{h,\alpha}^2 \quad (21)$$

where $g = \frac{b}{2a}$ and $h = \frac{2a^2}{b}$, with a and b are the mean and variance of the SPE index.

4. Online reduced KPCA

In literature, there are many online kernel methods for fault detection such as online KPCA [21] and SVD-RKPCA [15]. The online KPCA method is performed in two phases. The offline phase consists in determining the initial KPCA model and the SPE index and its control limit. In the online phase, we update only the SPE index. The number of kernel functions in the online KPCA method grows

linearly with the number of training data. The large number of kernel functions makes the kernel algorithm unsuitable for online and real time applications. On one hand, it makes the algorithm in danger of over-fitting when the number of kernel functions is too large. On the other hand, the computational complexity grows superlinearly with the number of kernel functions, which leads to a large training time for the update of the algorithm. To solve this problem, a dictionary containing the most relevant independent kernel functions is defined. For this purpose, an approximation criterion is developed to build the new reduced KPCA model and use it for process monitoring.

4.1. Approximation criterion

The (linear) approximation criterion [11–13] determines the relevance of discarding or accepting the kernel function $\phi(x_k)$ of the measurement x_k at time k in the dictionary $\mathcal{D}_r$. where $\mathcal{D}_r = \{\phi(x_{w_1}), \phi(x_{w_2}), \dots, \phi(x_{w_r})\}$, with r is the number of reduced data set and $\{w_1, w_2, \dots, w_r\} \subset \{1, 2, \dots, N\}$, is the dictionary. The dictionary $\mathcal{D}_r$ contains the most relevant r kernel functions among the N available. The approximation criterion consists of comparing $\phi(x_k)$ to its projection onto the principal component space spanned by the r elements of the dictionary. $\phi(x_k)$ is included in the dictionary if the following condition is satisfied:

$$\varepsilon_k = \min_{\beta} \|\phi(x_k) - \sum_{j=1}^r \beta_j \phi(x_{w_j})\|_H^2 \geq \nu \quad (22)$$

where ν is a positive threshold parameter that controls the level of sparsity. Note that Eq. (22) ensures the linear independence of the elements of the dictionary $\mathcal{D}_r$. Simplifying the left part of Eq. (22), we get:

$$\begin{cases} \varepsilon_k = \min_{\beta} \sum_{j,i=1}^r \beta_j \beta_i k(x_{w_j}, x_{w_i}) - 2 \sum_{j=1}^r \beta_j k(x_{w_j}, x_k) + k(x_k, x_k) \\ = \min_{\beta} \beta^T K_r \beta - 2 \beta^T k^r(x_k) + k(x_k, x_k) \end{cases} \quad (23)$$

where $K_r \in \mathbf{R}^{r \times r}$ is the reduced Gram matrix, $k^r(x_k)$ is:

$$k^r(x_k) = [k(x_{w_1}, x_k) \ k(x_{w_2}, x_k) \ \dots \ k(x_{w_r}, x_k)]^T \quad (24)$$

and β is the vector of optimal coefficients taking partial derivatives of Eq. (23) with respect to $\beta_1, \beta_2, \dots, \beta_r$, respectively, and let them be zero, which leads to:

$$\beta_k = K_r^{-1} k^r(x_k), \quad k = 1, \dots, N \quad (25)$$

By substituting Eq. (25) into Eq. (23), the expression of ε_k can be written as:

$$\varepsilon_k = k(x_k, x_k) - k^r(x_k)^T \beta_k \quad (26)$$

The resulting dictionary $\mathcal{D}_r$, called ν -approximate, satisfies the following relation [13,12]:

$$\min_{i=1, \dots, r} \min_{\beta_1, \dots, \beta_N} \|\phi(x_{w_i}) - \sum_{j=1}^r \beta_j \phi(x_{w_j})\| \geq \sqrt{\nu} \quad (27)$$

4.2. Online monitoring using reduced KPCA

In this section, online monitoring using reduced KPCA algorithm is derived. The parameters of the algorithm are updated only when the dictionary changes. This includes the update of the kernel matrix, its inverse, the number of principal components, the SPE index and its control limit. For a new observation x_k at instant k , an approximation criterion and evaluation of SPE index determine whether the dictionary remains unchanged or should be updated.

Indeed, the dictionary is updated if the new kernel function $\phi(x_k)$ is added to the dictionary since it cannot be well approximated by a linear combination of the dictionary elements and when the observation x_k is not faulty. For normal process change ($SPE_k^r < SPE_{\alpha, k-1}^r$), two cases may arise depending on the condition as in Eq. (22):

4.3. Case 1: unchanged dictionary ($\varepsilon_k < \nu$)

In this case, the new kernel function $\phi(x_k)$ is not added to the dictionary which remains unchanged. Using the dictionary $\mathcal{D}_r$, the expression of the k th eigenvector V_k^r in Eq. (8) can be written as:

$$V_k^r = \sum_{i=1}^r \alpha_{k,i}^r \phi(x_{w_i}), \quad k = 1, \dots, N \quad (28)$$

The kernel vector $k^r(x_k)$ in this case is given by:

$$k^r(x_k) = [k(x_{w_1}, x_k) \ k(x_{w_2}, x_k) \ \dots \ k(x_{w_r}, x_k)]^T \quad (29)$$

4.4. Case 2: changed dictionary ($\varepsilon_k \geq \nu$)

In this case, the kernel function $\phi(x_k)$ is added to the dictionary $\mathcal{D}_r$. So, the new dictionary becomes:

$$\mathcal{D}_r = [\phi(x_{w_1}) \ \phi(x_{w_2}) \ \dots \ \phi(x_{w_r}) \ \phi(x_{w_{r+1}})]^T \quad (30)$$

The j th eigenvector at instant k is given by:

$$V_k^r = \sum_{i=1}^{r+1} \alpha_{k,i}^r \phi(x_{w_i}), \quad k = 1, \dots, N \quad (31)$$

where $w_{r+1} = k$. In this case, the reduced kernel matrix $K_{r+1} \in \mathbf{R}^{(r+1) \times (r+1)}$ is updated as follows:

$$K_{r+1} = \begin{bmatrix} K_r & k^r(x_k) \\ k^r(x_k)^T & k(x_k, x_k) \end{bmatrix} \quad (32)$$

Similarly to the demonstration in Eq. (20), the SPE^r for an observation x_k using the dictionary $\mathcal{D}_r$ is defined as:

$$SPE^r = k(x_k, x_k) - (k^r(x_k))^T C^r (k^r(x_k)) \quad (33)$$

where,

$$\begin{cases} k^r(x_k) = [k(x_{w_1}, x_k) \ k(x_{w_2}, x_k) \ \dots \ k(x_{w_{r+1}}, x_k)]^T \\ C^r = P^r (\Lambda^r)^{-1} (P^r)^T \\ \Lambda^r = \text{diag}(\lambda_{\ell_1} \dots \lambda_{\ell_{r+1}}) \\ P^r = [\bar{\alpha}^r_{\ell_1} \ \bar{\alpha}^r_{\ell_2} \ \dots \ \bar{\alpha}^r_{\ell_{r+1}}] \\ \ell_{r+1} \text{ is the number of principal components} \\ \text{using the dictionary } \mathcal{D}_r \end{cases} \quad (34)$$

Note that when introducing a new observation in the dictionary, to update the vector of the coefficients β , the inverse of the kernel matrix K_{r+1}^{-1} is calculated. But, the calculation of this inverse can increase the computational complexity. To overcome this problem, the inverse of the kernel matrix is updated iteratively using the Woodbury matrix identity:

$$\begin{aligned} K_{r+1}^{-1} &= \begin{bmatrix} K_r^{-1} & 0 \\ 0 & 0 \\ K_r^{-1} & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{\varepsilon_k} \begin{bmatrix} -K_r^{-1} k^r(x_k) \\ 1 \\ -\beta_k \\ 1 \end{bmatrix} \begin{bmatrix} -k^r(x_k)^T K_r^{-1} & 1 \\ -\beta_k^T & 1 \end{bmatrix} \\ &= \begin{bmatrix} K_r^{-1} & 0 \\ 0 & 0 \\ K_r^{-1} & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{\varepsilon_k} \begin{bmatrix} -\beta_k \\ 1 \end{bmatrix} \begin{bmatrix} -\beta_k^T & 1 \end{bmatrix} \end{aligned} \quad (35)$$

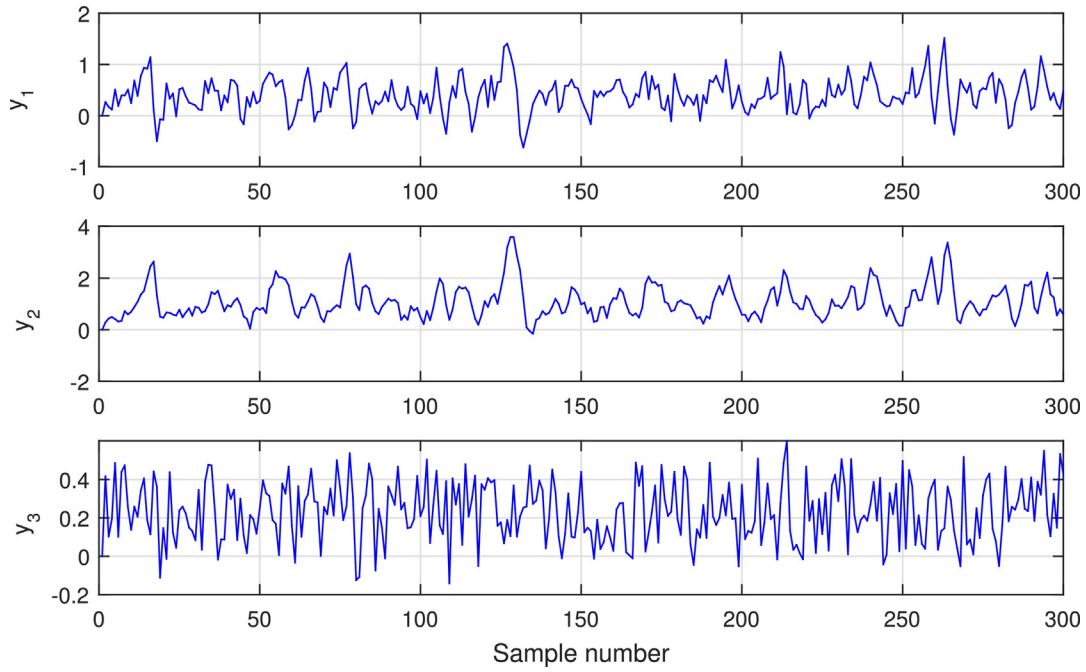


Fig. 1. Time evolution of the simulated data.

Thus, the vector of the coefficients β is updated as follows:

$$\beta_k = K_{r+1}^{-1} k^r(x_k) \quad (36)$$

In case of abnormal process changes ($SPE_k^r > SPE_{\alpha, k-1}^r$), the dictionary $\mathcal{D}_r$ remains unchanged, $\mathcal{D}_r = [\phi(x_{w_1}) \ \phi(x_{w_2}) \ \dots \ \phi(x_{w_r})]^T$. The algorithm which studies the developed online reduced KPCA is presented in Algorithm 1.

Algorithm 1. Online reduced KPCA algorithm

- 1: Initialization: $k = 1$, $r = 1$, SPE_k^r index, the control limit $SPE_{\alpha, k}^r$, the initial eigenvalue $\lambda_{\ell_{r,k}} = 1$, $w_k = 1$, $\ell_{r,k} = 1$, $K_k = k(x_k, x_k)$, $\beta_k = 1$,
- 2: $k \leftarrow k + 1$,
- 3: Obtain a new normal sample x_k and scale it,
- 4: Determine $k^r(x_k) = [k(x_{w_1}, x_k) \ \dots \ k(x_{w_r}, x_k)]^T$ and scale it,
- 5: Compute the monitoring index SPE_k^r for the new data x_k ,
- 6: **while** $SPE_k^r < SPE_{\alpha, k-1}^r$ **do**
- 7: $\beta_k \leftarrow K_r^{-1} k^r(x_k)$,
- 8: $\varepsilon_k \leftarrow k(x_k, x_k) - k^r(x_k)^T \beta_k$,
- 9: **if** $\varepsilon_k \geq \nu$ **then**,
- 10: $r \leftarrow r + 1$ and $w_r \leftarrow k$,
- 11: Update the kernel matrix K_{r+1} and its inverse K_{r+1}^{-1} then scale them,
- 12: Update $\ell_{r,k}$, V_k^r , $SPE_{\alpha, k-1}^r$
- 13: **else** dictionary remain unchanged
- 14: **end if**
- 15: **end while** Go to 2

5. Simulation results

In this section, two examples are presented in order to evaluate the performance of the online reduced KPCA method. The first one is a numerical example and the second one is a Tennessee Eastman Process which has been used widely for testing of various monitoring techniques. The detection performance of the developed online reduced KPCA is evaluated in terms of the following indicators:

- Good Detection Rate (GDR)(%): percentage of faulty observations detected.
- False Alarm Rate (FAR) (%): percentage of wrong fault declared in fault free region.
- Elapsed Time (ET) (seconds): measure of the real execution time.

5.1. Numerical example

In this section, the fault detection performance of the developed online reduced KPCA technique will be evaluated using three fault detection criteria: GDR, FAR and ET through a numerical example. In order to thoroughly assess the performance of the proposed online reduced KPCA, a comparison will also be made to the online KPCA. The data in this example are generated from the following nonlinear dynamic process that contains 3 variables [19,2,5]:

$$\begin{aligned} z(k) &= Az(k-1) + Bf(u(k-1)) \\ y(k) &= z(k) + v(k) \end{aligned} \quad (37)$$

where u , y and z are respectively, input, output and state variables. v is a random output noise, of $v \sim \mathcal{N}(0, 0.5)$. The matrices A and B are given by:

$$A = \begin{bmatrix} 0.118 & -0.191 & 0 \\ 0.847 & 0.264 & 0.9 \\ 0.214 & -0.11 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0.05 & -0.1 \\ 0.05 & 0.05 \\ 0 & 0.05 \end{bmatrix}$$

The correlated input $u(k) = [u_1(k) \ u_2(k)]^T$ is given by:

$$u(k) = \begin{bmatrix} 0.811 & -0.226 \\ 0.477 & 0.415 \end{bmatrix} u(k-1) + \begin{bmatrix} 0.193 & 0.689 \\ -0.320 & -0.749 \end{bmatrix} w(k-1)$$

with w is a random noise of $N(0, 5)$. The nonlinear function vector:

$$f(u(k-1)) = [u_1(k-1)^2 \ u_2(k-1)^2]^T$$

To build the KPCA model, 3 variables and 300 samples are used to form the training data. Fig. 1 shows response of the simulated example.

In the current work, the cumulative percent variance (CPV) is used to compute the number of important principal components. In this example, the kernel parameter of the RBF function is set to be $\sigma = 5.3$, $\nu = 0.05$ and SPE_{α} limit is computed with a confidence level $(1 - \alpha) = 0.95$. Using the proposed online reduced KPCA method, Fig. 2 shows the variation of the reduced number over time and Fig. 3 presents the number of principal components variation over

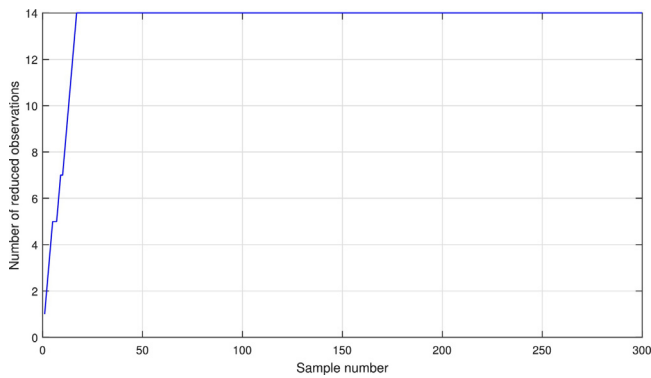


Fig. 2. Variation of the reduced number of observations.

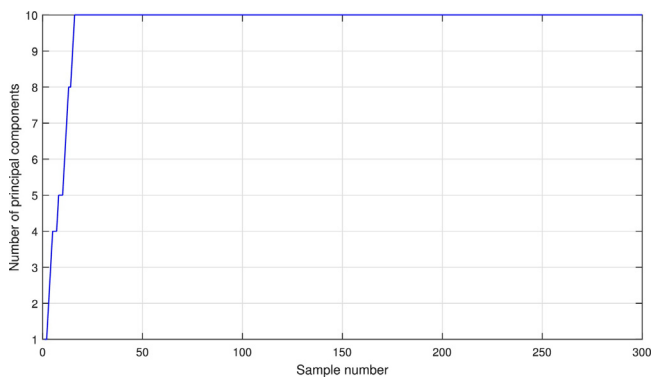


Fig. 3. Variation of the number of principal components.

time. As shown in these figures, for a data set of 300 observations, the number of reduced observations is equal to 14.

The monitoring results under normal operation using the SPE-based online KPCA and SPE^r -based online reduced KPCA techniques are shown in Figs. 4 and 5, respectively. In the normal conditions, the proposed technique (Fig. 5) gives better results when compared to the online KPCA technique (Fig. 4) which results a high false alarm rate.

The detection results in the training phase are summarized in Table 1 in terms of false alarm rate and elapsed time. The proposed method has less false alarm rate (0.67%) compared to the KPCA technique (4%). Using the proposed method, the elapsed time is reduced to 94.9647% by using a reduced number of kernel functions which is equivalent to 4.66% of the number of initial data.

Now, a bias fault is introduced in the variable x_2 with a magnitude equals to 38% of the variation range of the corresponding

Table 1
Summary of false alarm rates and elapsed time.

	Online reduced KPCA	Online KPCA
FAR (%)	0.67	4
ET (seconds)	0.093276	1.852458

Table 2
Summary of good detection rates, false alarm rates, and elapsed time for numerical data.

	Online reduced KPCA	Online KPCA
FAR (%)	1.34	13.37
GDR (%)	71.29	53.47
ET (seconds)	0.108907	2.822250

variable between samples 150 and 250. Figs. 6 and 7 show the fault detection results of the online KPCA and online reduced KPCA techniques. As can be seen from Fig. 6, the developed technique can detect the fault in the x_2 between the sample 150 and 250 and provides a good results compared to the online KPCA technique.

To further demonstrate the effectiveness of the proposed method under normal operation, we compare the elapsed time at each iteration for both online KPCA and online reduced KPCA methods. As seen from Fig. 8, the elapsed time of the proposed method is very small compared to the online KPCA.

In order to ensure that meaningful comparison can be made, FAR, GDR and ET metrics are computed for both online KPCA and online reduced KPCA methods. The fault detection results are summarized in Table 2. These results show that the proposed online reduced KPCA method reduces significantly both the FAR and ET and improves the GDR.

5.2. Tennessee Eastman Process

The TEP is a benchmark simulation model of a complex industrial chemical process. It was proposed by Downs and Vogel [7]. The TEP contains eight components: G and H are the main products, A , C , D and E are reactants, F is a byproduct and B is an inert component. The reactions that take place in the reactor are:



This process contains five major units: a reactor, a stripper, a condenser, a recycle compressor, and a separator. The TEP contains 41 measured variables (19 composition measurements and

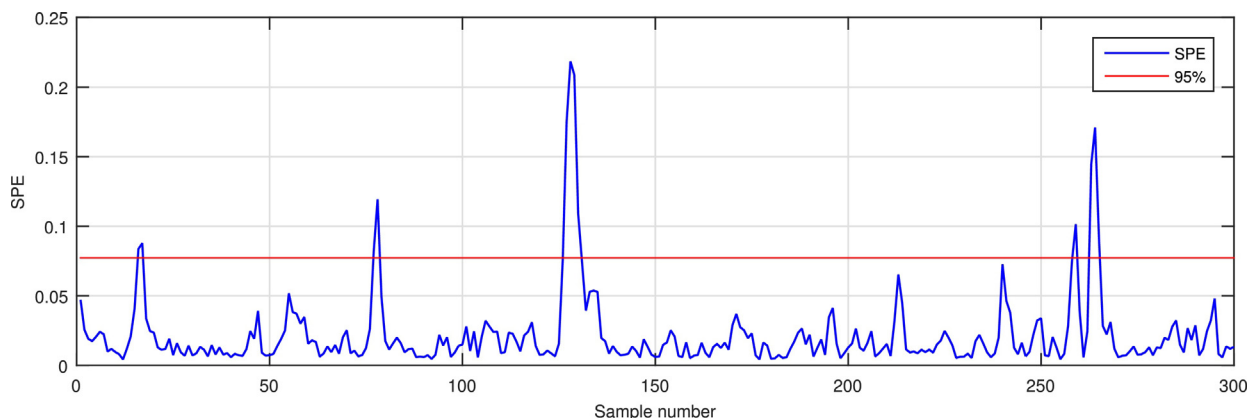


Fig. 4. Time evolution of SPE index under normal operation.

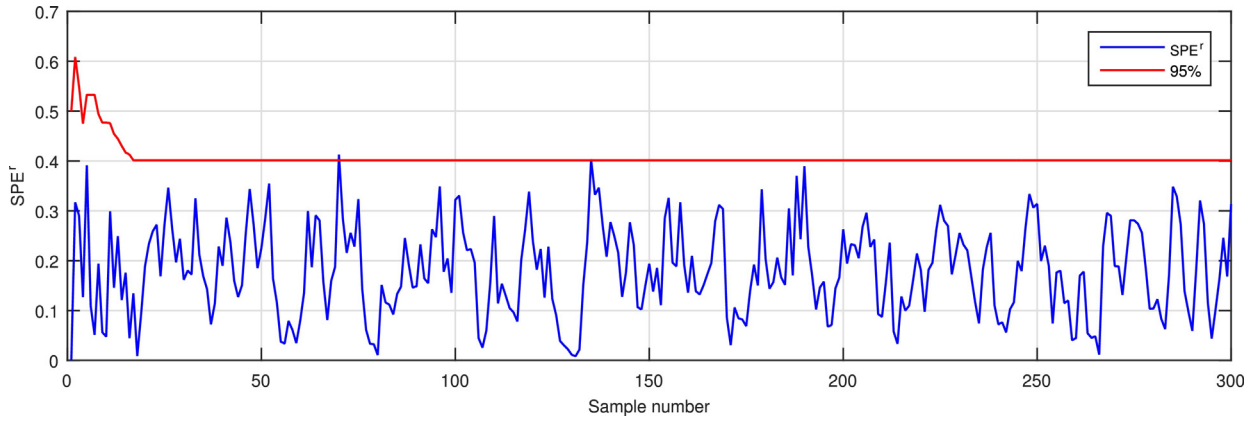


Fig. 5. Time evolution of SPE^r index under normal operation.

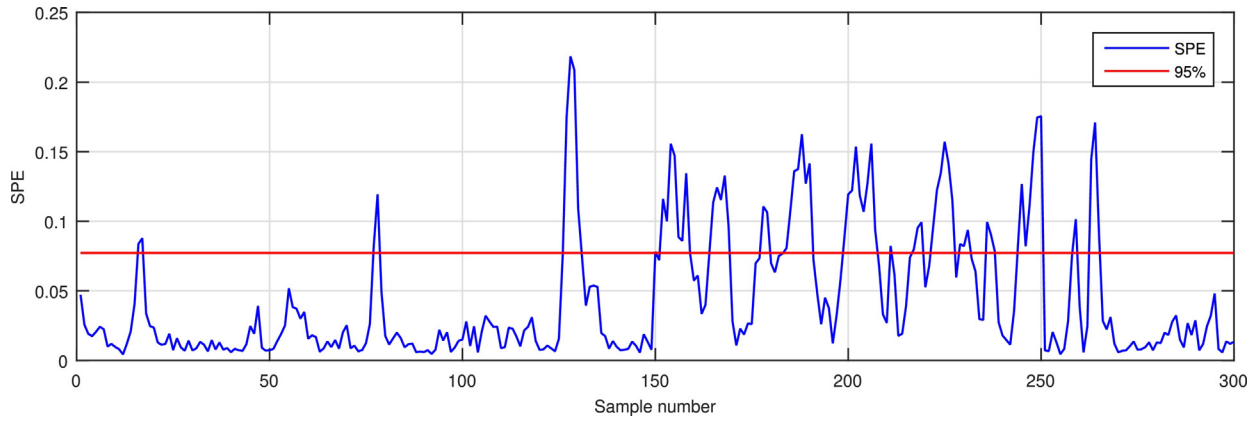


Fig. 6. Time evolution of SPE index with a fault on the variable x_2 .

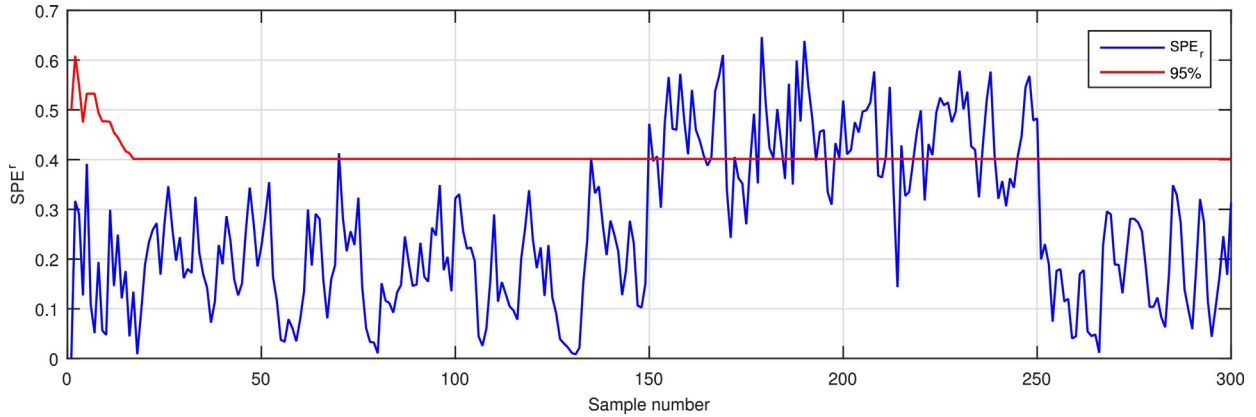


Fig. 7. Time evolution of SPE^r index with a fault on the variable x_2 .

22 continuous process measurements) and 12 manipulated variables. Details of TEP including the reactions are widely detailed in literature [7] (Fig. 9). 20 variables are considered for TEP monitoring and these variables are listed in Table 3.

In this study, 1000 observations are generated for process monitoring. The kernel parameter of the RBF function is set to 32. The number of kernel principal components is chosen using the CPV method, such that the first eigenvalues represent about 95% of the total variance. The confidence level is equal to 95%. The developed online reduced KPCA algorithm is applied to TEP to determine the reduced number of observations for monitoring. Figs. 10 and 11 present, respectively, the variation of the reduced number of

observations as well as the number of principal components under normal operation using the proposed method. We can show from these figures that the number of observations is reduced to 48 ($\nu = 0.18$).

The monitoring results under normal operation using both online reduced KPCA and online KPCA methods are presented in Figs. 12 and 13. We can see from these figures that the online reduced KPCA method (Fig. 13) provides further reduction in false alarm rate compared to the online KPCA (Fig. 12), which can be explained by the fact that the online KPCA model is not able to model the dynamic TEP efficiently.

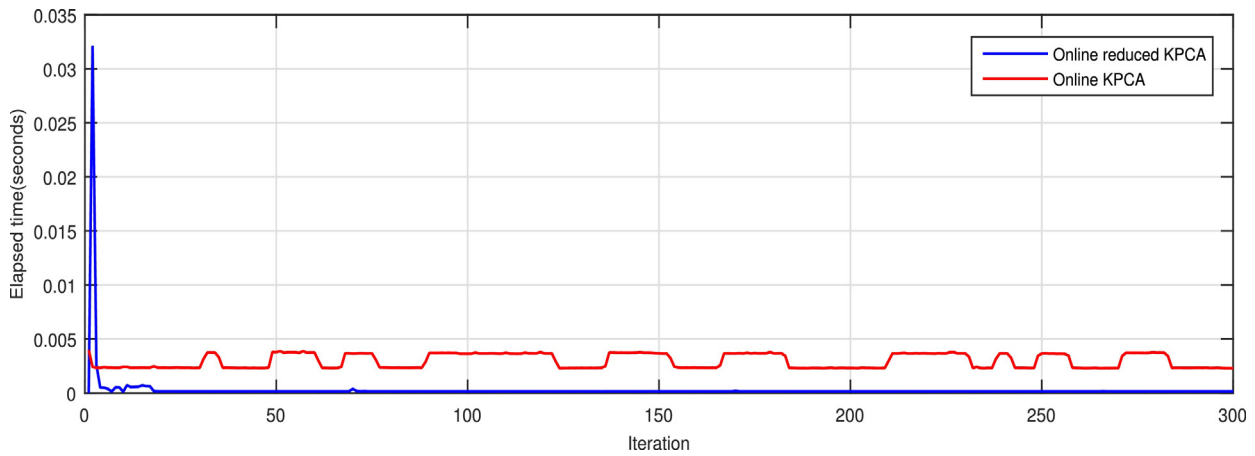


Fig. 8. Variation of the elapsed time.

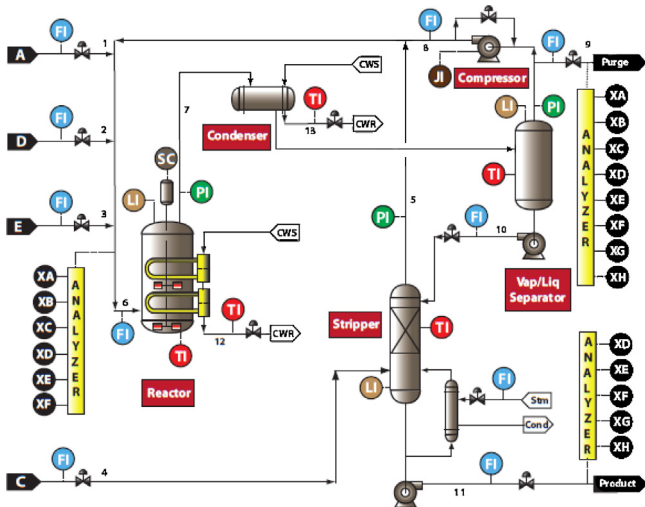


Fig. 9. Tennessee Eastman Process flow diagram.

Table 3
List of the monitoring variables in the TEP.

Variables	Measured variables	Unit
x_1	A feed (stream 1)	km^3/h
x_2	D feed (stream 1)	kg/h
x_3	E feed (stream 1)	kg/h
x_4	Total feed (stream 4)	km^3/h
x_5	Recycle flow (stream 8)	km^3/h
x_6	Reactor feed rate (stream 6)	km^3/h
x_7	Reactor pressure	kPa
x_8	Reactor level	%
x_9	Reactor temperature	$^{\circ}\text{C}$
x_{10}	Purge rate (stream 9)	km^3/h
x_{11}	Product separator temperature	$^{\circ}\text{C}$
x_{12}	Product separator level	%
x_{13}	Product separator pressure	kPa
x_{14}	Product separator underflow (stream 10)	m^3/h
x_{15}	Stripper level	%
x_{16}	Stripper pressure	kPa
x_{17}	Stripper underflow (stream 11)	m^3/h
x_{18}	Stripper temperature	$^{\circ}\text{C}$
x_{19}	Stripper stream flow	kg/h
x_{20}	Compressor work	kW

To further compare the performance of the online reduced KPCA and the online KPCA methods, the elapsed time at each iteration as well as the false alarm rates are computed. It should be noted, from Table 4 and Fig. 14, that the online KPCA method returns

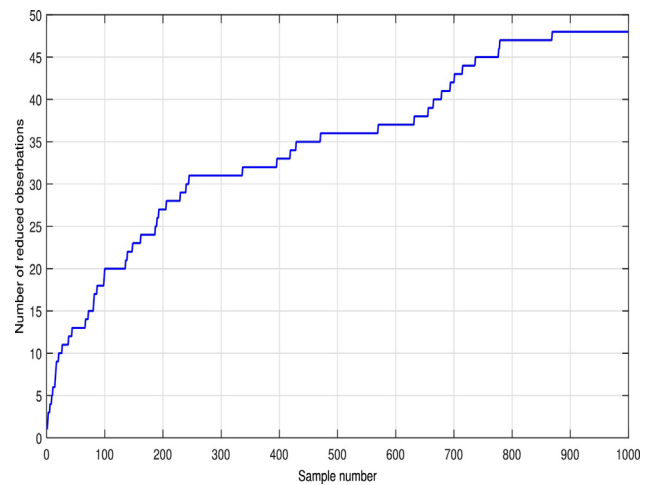


Fig. 10. Variation of the reduced number of observations.

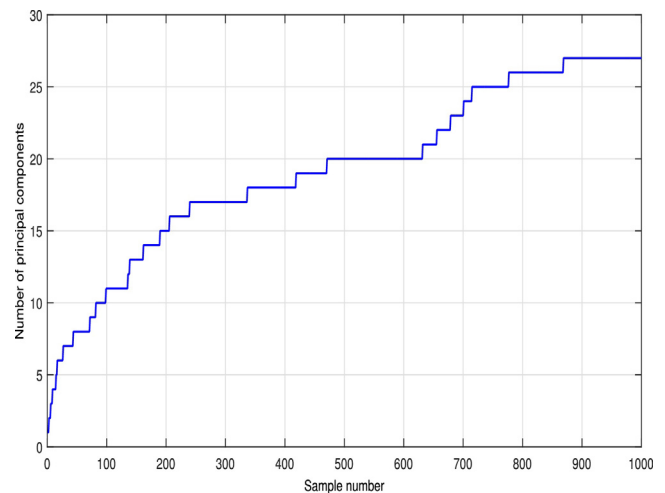


Fig. 11. Variation of the number of principal components.

Table 4
Summary of false alarm rates and elapsed time.

	Online reduced KPCA	Online KPCA
FAR (%)	0.3	3.4
ET (seconds)	0.471750	18.798056

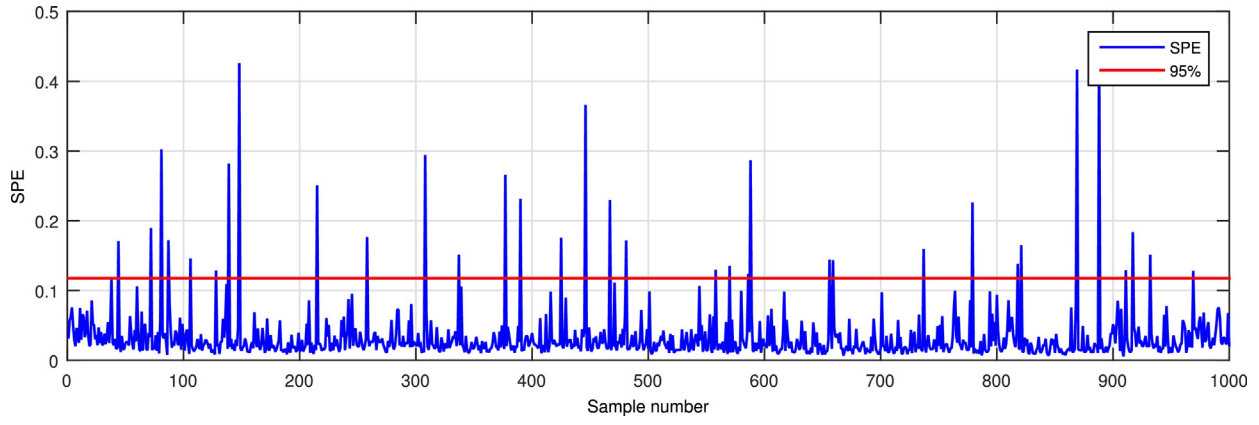


Fig. 12. Time evolution of SPE index under normal operation.

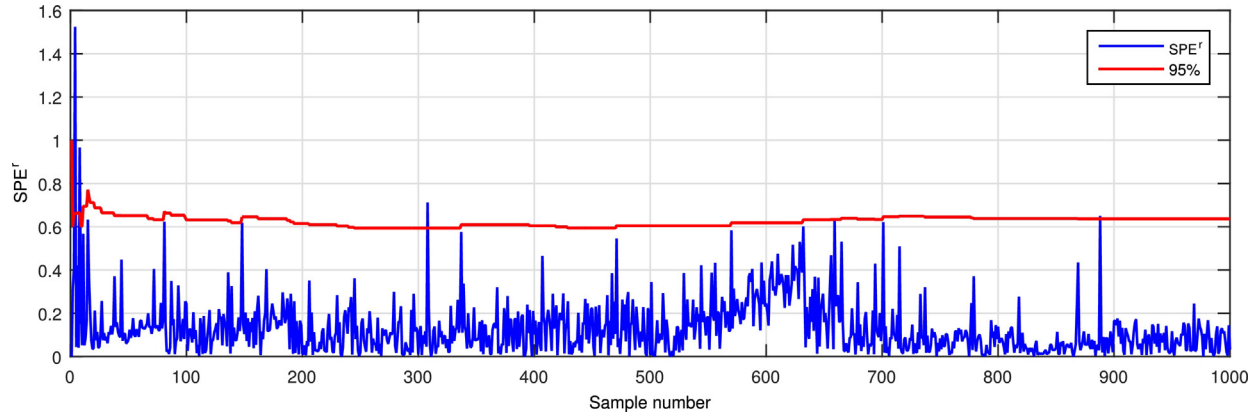


Fig. 13. Time evolution of SPE^r under normal operation.

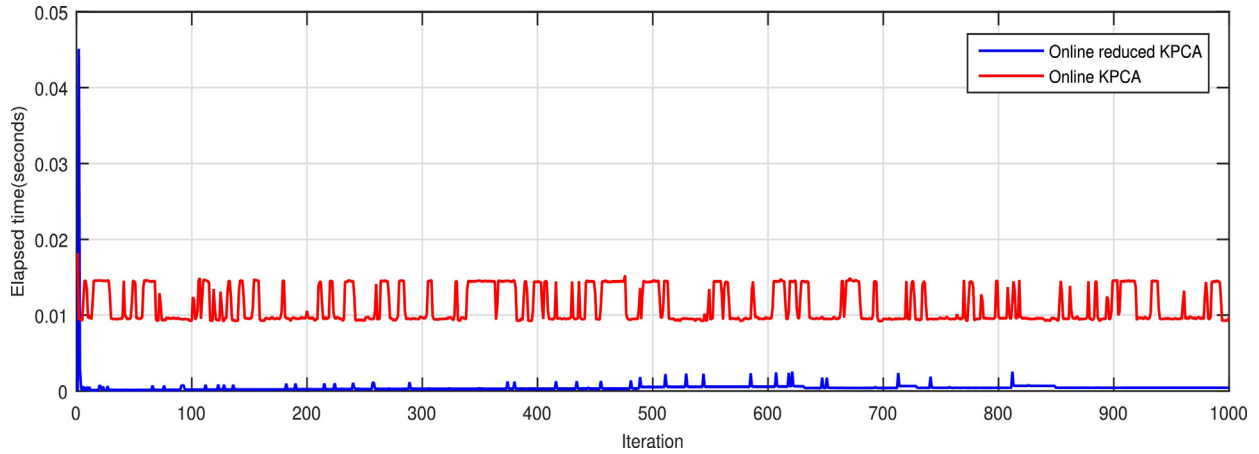


Fig. 14. Variation of the elapsed time.

higher elapsed time and false alarm rate compared to the online reduced KPCA. The elapsed time of the proposed method is reduced to 97.49% using a number of kernel functions which is equivalent to 4.9% of the number of initial data. Table 4 shows also that the online reduced KPCA method has less false alarm rate (0.5%) compared to the online KPCA method (3.4%).

Next, a bias fault is injected to the variable x_2 with an amplitude equal to 43% of the range of the variation of this variable between the instant 300 and 500. In this case, the variation of the reduced number of observations as well as the number of principal com-

ponents using the proposed method are presented respectively in Figs. 15.

In the presence of a bias fault in x_2 , we can show from Figs. 17 and 18 the monitoring results of TEP using the online KPCA (SPE) and the proposed online reduced KPCA (SPE^r) techniques. Both online KPCA and online reduced KPCA methods are able to detect the fault but with high false alarm and miss detection rates when using the online KPCA (Fig. 17), and 16

Table 5 shows the detection performances in terms of FAR, GDR and ET for both KPCA and online reduced KPCA methods. The results

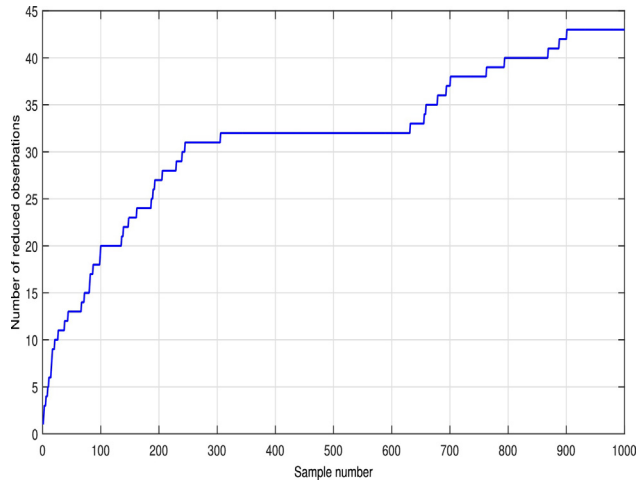


Fig. 15. Variation of the reduced number of observations.

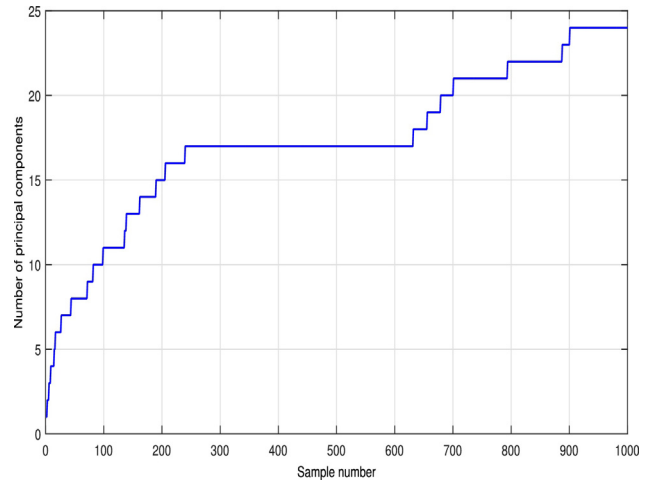


Fig. 16. Variation of the number of principal components.

Table 5
Summary of good detection rates, false alarm rates, and elapsed time for TEP data.

	Online reduced KPCA	Online KPCA
FAR (%)	1.47	6.54
GDR (%)	97.01	96.52
ET (seconds)	0.426162	30.338076

show that the ET required to perform fault detection with the online reduced KPCA method decreased because of the reduction of the size of the dictionary. The proposed online reduced KPCA provides a reduction of FAR (1.47%) and an increase of GDR (97.01%).

6. Conclusions

In this paper, an online reduced KPCA method is proposed for process monitoring to deal with the dynamic behaviors of the systems by changing the model structure. The proposed method consists in updating the KPCA model depending on the dictionary $\mathcal{D}_r$ which contains linearly independent kernel functions and then using this new reduced KPCA model for process monitoring. A comparison between the new online reduced KPCA method and the online KPCA method is done using a numerical example and the TEP in order to evaluate the performance of the proposed method.

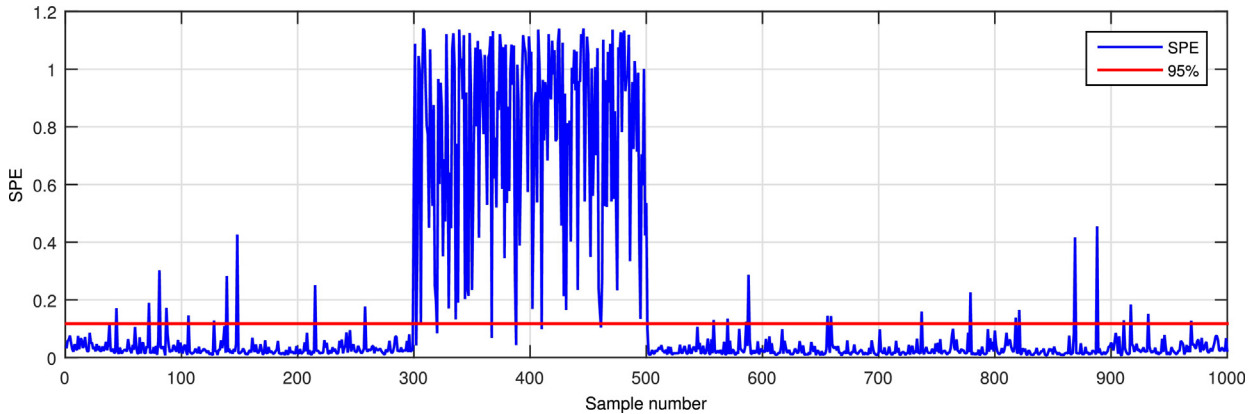


Fig. 17. Time evolution of SPE index with a fault on the variable x_2 .

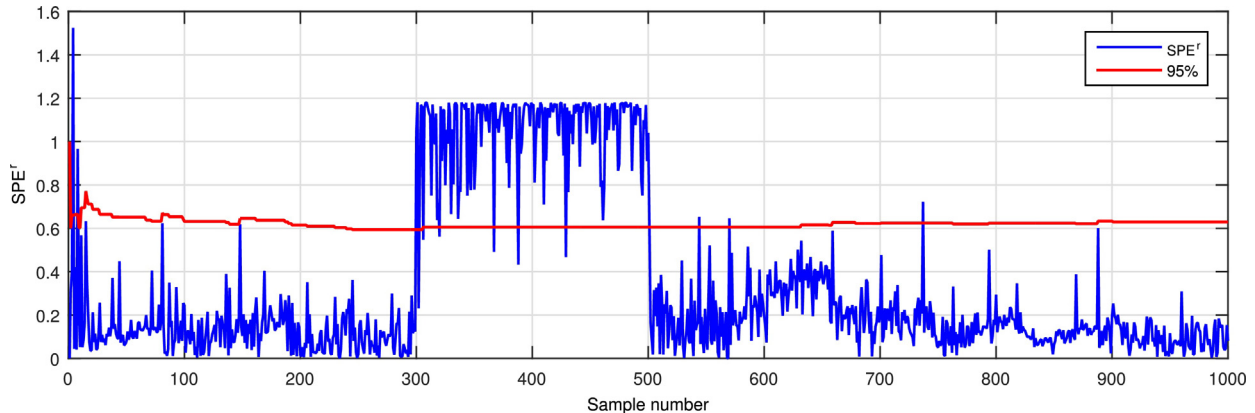


Fig. 18. Time evolution of SPE^r with a fault on the variable x_2 .

The simulation results show that the developed online reduced KPCA method outperforms the online KPCA method in terms of good detection rate, false alarm rate and elapsed time.

Acknowledgement

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